

CHE 313: HOMEWORK 6 RAJESH PILLAI
 PROBLEMS: 22.1-2, 22.1-4, 22.1-5, 22.1-7, 22.1-8, 22.5-3, 22.5-6

22.1-2 GAS SOLUBILITY IN AN AQUEOUS SOLUTION

20/ T = 303K $\rightarrow C_{CO_2} = .90 \times 10^{-4} \frac{kg CO_2}{kg H_2O}$ What P_{CO_2} must be kept to prevent CO_2 from vaporizing from solution?

20/ $H_{CO_2 \text{ in water}} = .186 \times 10^{-4} \frac{atm}{mol}$

$P_{CO_2} = H_{CO_2} \times C_{CO_2} = (.186 \times 10^{-4} \frac{atm}{mol}) (.90 \times 10^{-4} \frac{kg CO_2}{kg H_2O}) \times (\frac{18 \frac{kg H_2O}{kmol}}{44 \frac{kg CO_2}{kmol}}) \Rightarrow P_{CO_2} = (.1674 \frac{atm}{mol}) \times (\frac{18 \times 10^{-4}}{44} \frac{kmol}{kmol})$

$P_{CO_2} = .068451 \frac{atm}{mol} \times \frac{101325 Pa}{atm} \Rightarrow P_{CO_2} = 6938.92 Pa$

22.1-4 EQUILIBRIUM STAGE CONTACT FOR GAS-LIQUID SYSTEM

20/ Gas mixture @ $P = 2.026 \times 10^5 Pa$ contains air and SO_2 brought into contact in a SINGLE-STAGE EQUILIBRIUM MIXER w/ H_2O @ $T = 293K$ $x_{SO_2} = 0$
 $y_{P_{SO_2}} = P_{SO_2} = 1.52 \times 10^4 Pa$ INLET GAS: 5.70 total kmol INLET WATER: 2.20 total kmol

EXIT STREAMS in Equilibrium. Calculate outlet phases and compositions

① $L' \left(\frac{x_2}{1-x_2} \right) + V' \left(\frac{y_2}{1-y_2} \right) = L' \left(\frac{x_1}{1-x_1} \right) + V' \left(\frac{y_1}{1-y_1} \right)$
 $L_0 = L' = 2.2 \text{ kmol}$ $V' = V_2 (1 - y_2)$
 $V' = 5.70 (1 - 0.075) \rightarrow V' = 5.272$

$y_{A2} = \frac{P_{SO_2}}{P_{TOT}} = \frac{1.52 \times 10^4}{2.026 \times 10^5} \Rightarrow y_{A2} = .07503$

$x_{A1} = H_{A1} y_{A1} \rightarrow \text{From Fig. 22.1-1 } P_{A,SO_2} = H_{A1} x_{A1} = 29.6 x_{A1} \rightarrow x_{A1} = \frac{P_{A,SO_2}}{29.6} = \frac{1.52 \times 10^4}{29.6} = 513.51$

$x_{A1} P_{TOT} = 29.6 x_{A1} \rightarrow y_{A1} (2.026 \times 10^5) = 29.6 x_{A1} \rightarrow y_{A1} = 14.8 x_{A1}$ $x_{A1} = \frac{y_{A1}}{14.8}$

$2.2 \left(\frac{0}{1-0} \right) + 5.272 \left(\frac{0.07503}{1-0.07503} \right) = 2.2 \left(\frac{x_{A1}}{1-x_{A1}} \right) + 5.272 \left(\frac{y_{A1}}{1-y_{A1}} \right)$

$.4276406715 = 2.2 \left(\frac{x_{A1}}{1-x_{A1}} \right) + 5.272 \left(\frac{14.8 x_{A1}}{1-14.8 x_{A1}} \right) = \frac{2.2 x_{A1}}{1-x_{A1}} + \frac{78.031 x_{A1}}{1-14.8 x_{A1}}$

$.4276406715 (1-x_{A1})(1-14.8 x_{A1}) = 2.2 x_{A1} (1-14.8 x_{A1}) + 78.031 x_{A1} (1-x_{A1})$

$.4276 (1-15.4 x_{A1} + 14.8 x_{A1}^2) = 2.2 x_{A1} - 32.56 x_{A1}^2 + 78.031 x_{A1} - 78.031 x_{A1}^2$

$.4276 - 6.7567 x_{A1} + 6.3291 x_{A1}^2 = -110.591 x_{A1}^2 + 80.231 x_{A1} \rightarrow 116.92 x_{A1}^2 - 86.9876 x_{A1} + .4276 = 0$

$x_{A1} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \rightarrow x_{A1} = .00495$ $y_{A1} = 14.8 x_{A1} = 14.8 (.00495) \rightarrow y_{A1} = .0732$

$L_1 = \frac{L'}{1-x_{A1}} \rightarrow \frac{2.2}{1-.00495} \rightarrow L_1 = 2.211 \text{ total kmol}$ $V_1 = \frac{V'}{1-y_{A1}} = \frac{5.2725}{1-.0732} \rightarrow V_1 = 5.689 \text{ total kmol}$

BLEM 22.1-5 : ABSORPTION IN A COUNTERCURRENT STAGE TOWER (2)

$$L_0 = 108 \frac{\text{kmol H}_2\text{O}}{\text{hr}}$$

$$Y_{A,11} = .01 \quad X_{A,0} = 0 \quad V_{A,11} = 30.0 \frac{\text{kmol}}{\text{hr}}$$

Entering Acetone : $Y_{A,11} V_{A,11} = .01(30.0) = .30 \frac{\text{kmol acetone}}{\text{hr}}$

Entering Air : $(1 - Y_{A,11}) V_{A,11} = (1 - .01)(30.0) = 29.7 \frac{\text{kmol air}}{\text{hr}}$

Acetone Leaving in V_1 : $(.10)(30) = .30 \frac{\text{kmol}}{\text{hr}}$

Acetone Leaving in L_N : $(.90)(30) = .27 \frac{\text{kmol}}{\text{hr}}$

$$V_1 = \text{entering Air} + \text{leaving acetone} = 29.7 + .03 \rightarrow V_1 = 29.73 \frac{\text{kmol}}{\text{hr}}$$

$$Y_{A,1} = \frac{.03}{29.73} \rightarrow Y_{A,1} = .001009017$$

$$L_N = \text{water in} + \text{Acetone out} = 108 + .27 \rightarrow L_N = 108.27 \frac{\text{kmol}}{\text{hr}}$$

$$X_{A,N} = \frac{.27}{108.27} \rightarrow X_{A,N} = .0024937656$$

- Since the flow of liquid varies only slightly from $L_0 = 108$ to $L_N = 108.27$ and V from 30 to 29.73, the slope $\frac{L}{V}$ of the operating line in Eq. 10.3-13 is essentially constant. ~~By using the average~~ $\text{slope} = \frac{.001009017 - .01}{0 - .0024937656} = 3.605$

$$Y_A = 3.605 X_A \rightarrow \text{From the slope calculated}$$

it would take about 3.6 stages for this separation w/ a water inlet of

b) KREMSEER EQUATION $V_1 = 29.73 \frac{\text{kmol}}{\text{hr}} \quad Y_{A,1} = .001009017 \quad L_0 = 108$
 $X_{A,0} = 0$

Equilibrium relation: $Y_A = 2.53 X_A$ where $m = 2.53$ then $A_1 = \frac{L}{mV} = \frac{L_0}{m_0 V_1}$

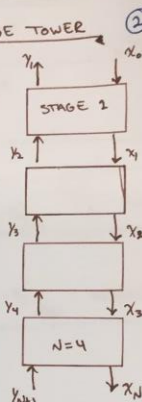
$$A_1 = \frac{108}{(2.53)(29.73)} \rightarrow A_1 = 1.435647529 \quad A_N = \frac{L_N}{m V_{A,11}} = \frac{108.27}{2.53(30)} \rightarrow A_N = 1.426442213$$

Geometric Avg: $\sqrt{A_1 A_N} = \sqrt{(1.436)(1.426)} \rightarrow A = 1.43115721$

$$22.1-28: N = \frac{\ln \left(\frac{Y_{A,11} - m X_{A,0}}{Y_1 - m X_0} \left(1 - \frac{1}{A} \right) + \frac{1}{A} \right)}{\ln A} \rightarrow N = \frac{\ln \left(\frac{.01 - 0}{.001009017 - 0} \left(1 - \frac{1}{1.4311} \right) + \frac{1}{1.4311} \right)}{\ln (1.4311)}$$

$$\rightarrow N = 3.638 \text{ stages}$$

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PROBLEM 22.1-7 | OVERALL MASS TRANSFER COEFFICIENT
FROM FILM COEFFICIENTS

FIGURE 22.1-9: $x_A^* = .35 \rightarrow$ in equilibrium w/ bulk gas $y_{AG} = .38$

$$m'' = \frac{y_{AG} - y_{Ai}}{x_A^* - x_{Ai}} = \frac{.38 - .197}{.35 - .257} \rightarrow m'' = 1.967741935$$

$$\frac{K_y}{(1-y_A)_{im}} = \frac{1.465E^{-3}}{.709} = .0020662906$$

$$\frac{1}{K'_x/(1-x_A)_{im}} = \frac{1}{m''K_y/(1-x_A)_{im}} + \frac{1}{k_x/(1-x_A)_{im}}$$

$$(1-x_A)_{im} = \frac{(1-x_{AL}) - (1-x_{AG})}{\ln \left[\frac{(1-x_{AL})}{(1-x_{AG})} \right]} = \frac{(1-.1) - (1-.35)}{\ln \left[\frac{(1-.1)}{(1-.35)} \right]} \rightarrow (1-x_A)_{im} = .7682323026$$

$$\frac{K_x}{(1-x_A)_{im}} = \frac{1.967E^{-3}}{.820} \rightarrow \frac{K_x}{(1-x_A)_{im}} = .0023987805$$

$$\frac{1}{K'_x/(1-x_A)_{im}} = \frac{1}{m''K_y/(1-x_A)_{im}} + \frac{1}{k_x/(1-x_A)_{im}} \rightarrow \frac{1}{K'_x/(.768)} = \frac{1}{(1.967)(.002)} + \frac{1}{(.0024)}$$

$$\rightarrow \frac{.768232026}{K'_x} = 245.9463937 + 416.8784931 \rightarrow K'_x(662.8248869) = .768232026$$

$$K'_x = .0011590275 = \frac{K_x}{(1-x_A)_{im}} \rightarrow K_x = \frac{.0011590275}{.7682323026} \rightarrow K_x = .001508694$$

$$K_x = .001508694 = 1.509E^{-3}$$

FLUX AND RESISTANCE: $\frac{1}{K'_x} = \frac{(1-x_A)_{im}}{m''K_y} + \frac{(1-x_A)_{im}}{k_x} = \frac{1}{\frac{(1.967)(1.465E^{-3})}{.709}} + \frac{1}{\frac{(1.46E^{-3})}{.820}}$

$$\Rightarrow 245.9463997 + 416.8784952 = 662.8248949$$

$$\% \text{ RESISTANCE} = \frac{\text{gas film resist}}{\text{total resist}} = \frac{245.95}{662.82} \rightarrow$$

$$\% \text{ RESISTANCE} = .37106 = 37.11\%$$

$$N_A = \frac{k'_x}{(1-x_A)_{im}} (x_{Ai} - x_{AL}) = \frac{(1.967E^{-3})}{(.820)} (.257 - .10) \rightarrow N_A = 3.766E^{-4} \frac{\text{kmol}}{\text{m}^2 \cdot \text{s}}$$

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BLEM 22.1-8 INTERFACE CONCENTRATIONS AND OVERALL
MASS-TRANSFER COEFFICIENTS

(4)

Equilibrium Data + Film Coefficients $k'_x + k'_y$ From Example 22.1-5

Use Bulk Concentrations of $y_{A0} = .25$ $x_{A0} = .05$

(a) Interface Concentrations y_{Ai} , x_{Ai} , and Flux N_A

$$K_y = 1.465 \times 10^{-3} \quad \text{From graph} \quad y_A^* = .025 \quad x_A^* = .265$$

$$K_x = 1.967 \times 10^{-3} \quad y_{Ai} = .147 \quad x_{Ai} = .257$$

$$(1-y_A)_{im} = \frac{[(1-y_{Ai}) - (1-y_{A0})]}{\ln \left[\frac{(1-y_{Ai})}{(1-y_{A0})} \right]} = \frac{.703 \left[\frac{(1-.147) - (1-.25)}{(1-.25)} \right]}{\ln \left[\frac{(1-.147)}{(1-.25)} \right]} =$$

$$(1-x_A)_{im} = \frac{[(1-x_{Ai}) - (1-x_{A0})]}{\ln \left[\frac{(1-x_{Ai})}{(1-x_{A0})} \right]} = .920$$

$$\frac{-k'_x / (1-x_A)_{im}}{k'_y / (1-y_A)_{im}} = \frac{(-1.465 \times 10^{-3} / .920)}{(1.967 \times 10^{-3} / .703)} \rightarrow \text{slope} = -1.15927412$$

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22.5-3 %/20

22.5-6 %/20